NANYANG JUNIOR COLLEGE
Year 2 Preliminary Examination

H2 GEOGRAPHY 9173/01

Paper 1 10 Sept 2024

3 hours

Additional Materials: Insert

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, index number and name on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE ON ANY BARCODES.

Answer both questions in **Section A**.
Answer two questions in **Section B**, one from each cluster.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Answer **both** questions.

Cluster 1: Development, Economy and Environment

1 Resource 1 shows the growth in solar installed capacity In India from 2008 to June 2023 (in megawatts). Resource 2A is a satellite image of Kanmai 2 in Kamuthi, Tamil Nadu, before the construction of Kamuthi solar power plant in 2009. Resource 2B is a satellite image of Kanmai 2 in Kamuthi, Tamil Nadu, after the construction of Kamuthi solar power plant in 2020. Resource 3 is an excerpt taken from an article titled 'Adani's Kamuthi solar power plant casts shadow on livelihood of locals'.

(a) Cite data from Resource 1 to describe the growth in India's solar installed capacity from 2008 to June 2023. [3]

(b) Suggest possible reasons to explain the growth in India's solar installed capacity shown in Resource 1. [5]

(c) Describe the differences in land use in Kamuthi between 2009 and 2020 shown in Resources 2A and 2B respectively. [4]

(d) Explain how solar power plants can impact the environment in India. [6]

(e) With reference to Resources 2A and 3, describe the ecosystem services that Kamuthi can provide prior to the construction of the solar power plant. [6]

(f) With reference to Resource 3, explain the socio-economic trade-offs associated with solar power. [6]

Cluster 2: Tropical Environments

2 Resource 4 is a photograph of the karst landscape in Tam Coc National Park, Ninh Binh, Vietnam. Resource 5 shows the climograph of Ninh Binh in Vietnam. Resource 6 is a photograph of Thien Cung Cave, Ha Long Bay, Vietnam.

(a) Describe the characteristics of the karst landform A in Tam Coc National Park, Ninh Binh, Vietnam, shown in Resource 4. [4]

(b) With reference to Resource 5, explain how the karst landform A shown in Resource 4 is formed over time. [6]

(c) With reference to Resource 6, explain how feature B is formed. [4]

(d) With reference to Resources 4 and 6, explain how human activities can accelerate the weathering processes in the karst landscapes in Vietnam. [6]

(e) With reference to Resource 6, suggest the ecosystem services that may be provided by the cave. [4]

(f) Explain how human activities can affect ecosystem services provided by karst landforms. [6]

[Turn over

Section B

Answer **either** question 3 **or** question 4 and answer **either** question 5 **or** question 6.

Cluster 1 Development, Economy and Environment

3 Evaluate the success of strategies to manage impacts of extractive industries on places. [20]

4 'The Global Production Networks (GPNs) of Transnational Corporations (TNCs) positively impact host economies more than home economies.' Evaluate this statement. [20]

Cluster 2 Tropical Environments

5 'Synoptic scale circulation contributes to the distinctive rainfall patterns in the tropics.' Evaluate this statement. [20]

6 'Fluvial deposition has the greatest impact on fluvial landforms in the humid tropics.' Evaluate this statement. [20]

End of Paper